LOAD BALANCING BASED ON AI IN ULTRA-DENSIFICATION HETEROGENEOUS NETWORKS

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Abstract:

Multi-label emotion detection in social media dis- course is fundamentally challenged by extreme class imbalance, where ”long-tail” emotional states are frequently marginalized by majority-class dominance. This research proposes a synergistic framework to mitigate this imbalance using the GoEmotions dataset, a complex 28-label taxonomy of Reddit comments. While standard transformer-based models achieve high global accuracy, they often exhibit poor macro-recall for rare emotions. Our approach integrates a two-stage optimization pipeline: (1) a cost-sensitive training phase utilizing Weighted Binary Cross-Entropy (WBCE) to reshape the gradient landscape, and (2) a post-hoc decision calibration phase employing per-label threshold optimization. Experimental results demonstrate that this synergy effectively recovers minority-class signals that are typically lost in standard pipelines. The proposed model achieved a Macro-F1 score of 0.496, driven by a 62.7% relative increase in Macro- Recall (0.352 to 0.573) compared to a standard DistilBERT base- line. Ablation studies confirm that training-time weighting and inference-time calibration are mutually dependent; weighting en- sures feature representation while calibration stabilizes precision. This work provides a scalable methodology for equitable affective computing, ensuring robust performance across the full spectrum of human sentiment in resource-constrained environments.

PROBLEM STATEMENT: *The rapid growth of IoT/IoE and mobile traffic in ultra-dense heterogeneous networks creates significant challenges in resource allocation, energy efficiency, and real-time load balancing, particularly for latency-sensitive applications such as healthcare.*

RESEARCH GAP:

- Struggle with energy constraints and scalability
- Lack intelligent, adaptive decision-making
- Often ignore real-time big data integration and distributed learning
- Provide limited support for heterogeneous multi-network environments (5G/6G, IoT, WSNs)

DATASET USED:

Just name and details of dataset, just in one or two lines

KEY CONTRIBUTIONS:

- This paper proposes an AI-driven, energy-aware load balancing framework integrating:
- Big Data Analytics (BDA): For real-time monitoring, prediction, and decision-making
- CHROA (Chaotic Horse Ride Optimization Algorithm): For global optimization and efficient load distribution

PROPOSED METHODOLOGY:

- This paper proposes an AI-driven, energy-aware load balancing framework integrating:
- Big Data Analytics (BDA): For real-time monitoring, prediction, and decision-making
- CHROA (Chaotic Horse Ride Optimization Algorithm): For global optimization and efficient load distribution
- Cloud-enabled IoT architecture: Using Hadoop, Spark, and Flume for scalable data processing
- Software-defined edge computing + DRL (PPO): For resource scheduling modeled as an MDP
- Federated Learning (FL): For privacy-preserving traffic prediction

KEY RESULTS

- Network Lifetime: Improved up to 24,404 rounds (highest among all models)
- Energy Consumption: Reduced to 13.84 J (lowest among compared methods)
- Throughput: Achieved 77.43 Kbps, outperforming all baselines
- Normalized Overhead: Reduced to ~6.11% at high node density

CONCLUSION / IMPACT

The proposed CHROA-based framework provides a scalable, energy-efficient, and intelligent solution for load balancing in ultra-dense heterogeneous networks. It is particularly suitable for real-time IoT applications (e.g., healthcare) requiring low latency, high throughput, and sustainable performance.

THREE TO FOUR IMPORTANT GRAPHICAL RESULTS